Napalm: The Web Creation Kit

Project Team

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Contents

List of Figures

List of Tables

Chapter 1

Introduction

The requirements for Napalm, a web development tool that facilitates drag-and-drop website building, are described in the document. This document's target audience consists of developers, project managers, testers, documentation authors, and marketing groups. The project's goal is to make website development easier for non-technical people while giving developers access to cutting-edge features. With features like third-party deployment possibilities, AI-powered content creation, and integrated real-time collaboration, Napalm presents itself as a versatile and reliable substitute for current web development platforms such Webflow and Wix.

1.1 Problem Statement

Many freelancers, startups, and small enterprises require websites; however, current platforms such as Webflow and Wix impose restrictions, entangling users in proprietary ecosystems. These platforms frequently have poor AI capabilities, don't export production-ready code, and restrict customisation. Additionally, they restrict personalisation by not giving users the freedom to select AI models that are suited to their requirements. Napalm fills these gaps by providing an open-source, no-code platform that facilitates real-time collaboration, exportable, production-ready code, and customisable AI integrations. This removes the need for technical know-how or dependence on proprietary tools, giving people complete control over the process of creating websites.

1.2 Scope

Napalm's primary focus is on offering a user-friendly drag-and-drop website construction interface that employs artificial intelligence (AI) to generate content. Users can export production-ready code for external deployment or additional customisation. The software is appropriate for small organisations, agencies, freelancers, and individual users because it allows teams to collaborate in real time. Additionally, Napalm incorporates open-source AI models for the creation of customised text, graphics, and layouts. Additionally, it offers deployment choices on third-party platforms such as Netlify and Vercel, guaranteeing flexibility in the location and method of website hosting. Both developers and non-technical users are empowered by the platform's emphasis on customisation, scalability, and simplicity.

1.3 Modules

Napalm consists of six primary modules, each focusing on a specific set of functionalities:

1.3.1 Module 1: UI Design, Editor, and Templating Module

This module provides an intuitive design interface for website creation. Users can design websites through drag-and-drop functionality, using pre-built flexible templates.

1. Design the user interface for web platform.
2. Provide an intuitive editor with customizable website templates.

1.3.2 Module 2: AI Content Generation Module

This module integrates open-source AI models to assist in generating content for websites based on user input.

1. AI models for automatic content generation.
2. User-driven input to customize generated content.

1.3.3 Module 3: User Authentication, Management, and Database Module

This module manages secure user authentication, user profiles, and database interactions.

1. Implement user login and profile management.
2. Manage user-related data using MongoDB and Prisma.

1.3.4 Module 4: WebSocket Module

This module facilitates real-time collaboration, enabling users to work simultaneously on website creation.

1. Real-time content synchronization.
2. Live collaboration tools.

1.3.5 Module 5: Third-Party Deployment Module

This module supports the deployment of websites on platforms like Vercel and Netlify.

1. Configure deployment settings for external platforms.
2. Manage and publish websites to third-party services.

1.3.6 Module 6: Testing and Optimization Module

This module ensures the platform's performance and scalability.

1. Perform system testing for functionality.
2. Optimize platform performance and security.

1.4 User Classes and Characteristics

The following are the anticipated user classes for Napalm, each with specific characteristics:

User class	Description
User	Regular users of the platform who can create and manage their websites using templates provided by the system. They can select from pre-defined templates or start projects from scratch.
Admin	Administrators who manage platform settings, user access, and permissions. They are responsible for overseeing user projects, managing templates, and handling platform optimization and updates.
Project	This class represents an individual project, storing all necessary information related to the website, such as user settings, content, and design choices. Users can have multiple projects, each with unique configurations.
Template	This class holds pre-designed website templates that users can select when starting a new project. Templates provide a structure for website layout, which users can customize using the drag-and-drop editor.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter outlines the functional and non-functional requirements for Napalm. It includes a use-case/event-response storyboard, followed by detailed descriptions of the system's functional and non-functional requirements.

2.1 Use-Case Diagram

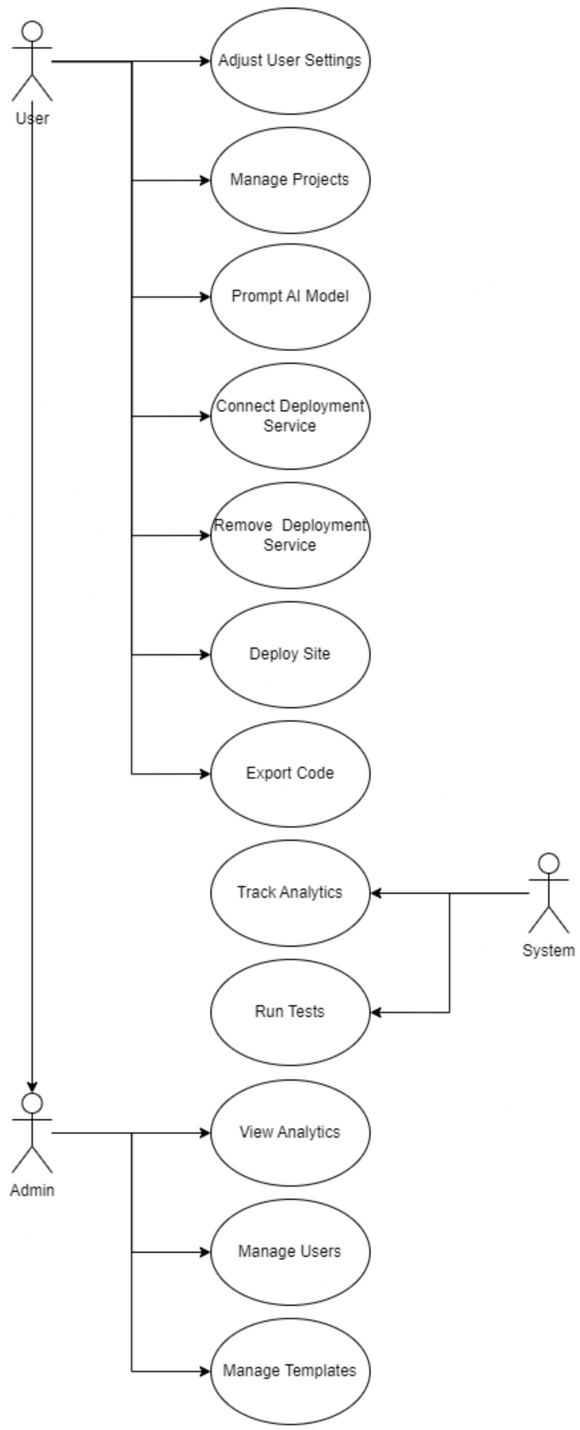


Figure 2.1: Use Case Diagram

2.2 Functional Requirements

The functional requirements of Napalm are described below, organized by modules. Each module represents a major component of the system.

2.2.1 Module 1: UI Design, Editor, and Templating Module

This module allows users to design and build websites using drag-and-drop functionality.

1. FR1: The system shall provide a user-friendly drag-and-drop interface for website creation.
2. FR2: Users shall be able to choose from a set of pre-designed templates.
3. FR3: The system shall allow users to edit and customize templates to suit their preferences.
4. FR4: The system shall store project files and design data in a format that can be exported as production-ready code.

2.2.2 Module 2: AI Content Generation Module

This module assists users in generating website content using AI.

1. FR1: The system shall integrate open-source AI models to generate content (text, images, etc.) based on user inputs.
2. FR2: Users shall have the option to customize the AI-generated content.
3. FR3: The system shall allow users to regenerate content or provide additional inputs to refine it.

2.2.3 Module 3: User Authentication, Management, and Database Module

This module handles secure user authentication and manages data storage.

1. FR1: The system shall provide secure login and user authentication.
2. FR2: Users shall have the ability to manage their profiles, including updating personal information.

3. FR3: The system shall store user and project data in a MongoDB database via Prisma.

2.2.4 Module 4: WebSocket Module

This module enables real-time collaboration between multiple users.

1. FR1: The system shall allow multiple users to work on the same project in real-time, with synchronized updates.
2. FR2: Users shall be able to see changes made by others in real-time without refreshing the page.

2.2.5 Module 5: Third-Party Deployment Module

This module provides functionality for deploying completed websites to third-party platforms.

1. FR1: The system shall allow users to deploy their website projects to Vercel and Netlify.
2. FR2: Users shall be able to configure deployment settings and manage the deployment process within the platform.

2.2.6 Module 6: Testing and Optimization Module

This module ensures that the system functions optimally and securely.

1. FR1: The system shall run automated tests to ensure that all functionalities work as expected.
2. FR2: The system shall provide performance optimizations to reduce page load times and enhance user experience.

2.3 Non-Functional Requirements

The non-functional requirements ensure that Napalm meets reliability, performance, usability, and security standards.

2.3.1 Reliability

1. NFR1: The system shall have an uptime of at least 99.9

2.3.2 Usability

1. USE-1: Users shall be able to create a new project within 5 minutes.

2. USE-2: The system shall allow users to customize a template with no more than 10 clicks.

2.3.3 Performance

1. PER-1: 95 percent of webpages generated by the platform shall load within 3 seconds on a standard 20 Mbps internet connection.

2. PER-2: The system shall support up to 5 users working simultaneously on the same project without performance degradation.

2.3.4 Security

1. SEC-1: Users' passwords shall be stored securely using hashing algorithms like bcrypt.

2. SEC-2: The system shall have protection against common web attacks like SQL injection and cross-site scripting (XSS).

2.4 Domain Model

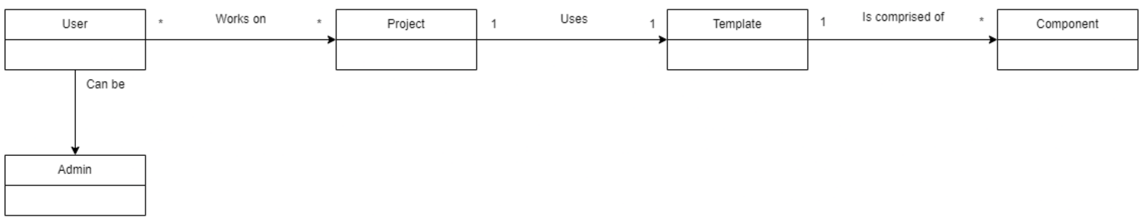


Figure 2.2: Domain Model

2.5 High-Level Use Cases

2.5.1 Adjust User Settings

Actors	User
Type	Primary
Description	The user can modify personal settings like profile information, password, and preferences. This allows the user to customize their experience based on their preferences.

2.5.2 Manage Projects

Actors	User
Type	Primary
Description	Users can create new projects, update existing ones, and delete projects they no longer need.

2.5.3 Prompt AI Model

Actors	User
Type	Primary
Description	Users can provide prompts to the AI model to generate content or designs for their projects. The AI will respond with edits which the user can review, undo, or keep.

2.5.4 Connect Deployment Service

Actors	User
Type	Primary
Description	The user selects and connects third-party deployment services like Vercel, Netlify, or other platforms to the system, allowing for continuous integration and deployment of their projects.

2.5.5 Remove Deployment Service

Actors	User
Type	Primary
Description	The user can disconnect a previously linked deployment service, ensuring that the system no longer pushes updates or deploys content to that platform.

2.5.6 Deploy Site

Actors	User
Type	Primary
Description	The user initiates the process to deploy a project (website) on the selected deployment platform. This action pushes the site to live servers, making it accessible publicly or within specified environments.

2.5.7 Export Code

Actors	User
Type	Primary
Description	Users can export the generated website code, enabling them to download the project for external usage. This can be useful for further customization, backups, or self-hosting.

2.5.8 Track Analytics

Actors	System
Type	Primary
Description	The system tracks template usage, engagement, and other key analytics for templates used in user projects.

2.5.9 Run Tests

Actors	System
Type	Primary
Description	The system automatically runs a series of tests (such as performance or functionality tests) on the Next.JS application. Ensures performance and reliability standards.

2.5.10 View Analytics

Actors	Admin
Type	Primary
Description	Admin users have access to advanced analytics for projects and websites, including user engagement, traffic trends, and system performance. They can analyze this data to make informed decisions about optimizations or updates.

2.5.11 Manage Users

Actors	Admin
Type	Primary
Description	Admins can manage the platform's user base by adding new users, editing user roles, suspending accounts, or deleting users. This provides administrative control over who can access the system and what actions they are permitted to take.

2.5.12 Manage Templates

Actors	Admin
Type	Primary
Description	Admins oversee the management of website templates, including creating new templates, updating existing ones, or removing outdated templates. These templates are made available to regular users for their projects.

2.6 Fully-Dressed Use Cases

Use Case Name	Adjust User Settings
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	User
Stakeholders and Interests	User: Modify personal settings to customize their experience System Admin: Ensure settings are stored correctly and securely
Preconditions	User is logged into the system, user profile data exists
Success Guarantee (Postconditions)	Settings are updated, saved, and applied successfully
Main Success Scenario	1. User navigates to settings page 2. System displays current settings 3. User updates settings 4. System validates and saves changes 5. Confirmation message is shown
Extensions	3a. Invalid input: System prompts to correct input 5a. Database error: System notifies and suggests retry
Special Requirements	Response time < 2 seconds, passwords encrypted

Table 2.1: Adjust User Settings

Use Case Name	Manage Projects
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	User
Stakeholders and Interests	User: Create, update, and delete projects as needed System Admin: Ensure project integrity and database stability
Preconditions	User is logged into the system and has the necessary permissions
Success Guarantee (Postconditions)	Projects are created, updated, or deleted successfully
Main Success Scenario	1. User navigates to Manage Projects page 2. System displays a list of existing projects 3. User creates, updates, or deletes a project 4. System saves and updates the project list
Extensions	3a. Missing required details: System prompts user to provide required fields 5a. Delete confirmation: System asks for confirmation before deleting
Special Requirements	Projects listed within 3 seconds, secure deletion of data

Table 2.2: Manage Projects

Use Case Name	Prompt AI Model
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	User
Stakeholders and Interests	User: Generate content using AI assistance to improve workflow System Admin: Manage AI model integration and ensure performance
Preconditions	User is logged in with an active project and AI model is available
Success Guarantee (Postconditions)	AI-generated content is displayed and usable by the user
Main Success Scenario	1. User selects AI prompt feature 2. System asks for user input or prompt 3. User submits prompt 4. System generates and displays content 5. User accepts or modifies content
Extensions	4a. AI fails to generate content: System shows error and suggests retry 6a. User modifies content: System saves modified version
Special Requirements	AI content generated in <10 seconds, user can refine content

Table 2.3: Prompt AI Model

Use Case Name	Connect Deployment Service
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	User
Stakeholders and Interests	User: Connect a third-party service to deploy their projects System Admin: Ensure deployment integrations function correctly
Preconditions	User is logged in and has an active project, deployment service is available
Success Guarantee (Postconditions)	Deployment service is successfully connected to the project
Main Success Scenario	1. User navigates to Deployment Settings 2. System displays available deployment services 3. User selects service and provides credentials 4. System validates and connects the service
Extensions	4a. Invalid credentials: System prompts for valid credentials 5a. Service unavailable: System notifies user and suggests retry
Special Requirements	Secure handling of credentials, connection process <5 seconds

Table 2.4: Connect Deployment Service

Use Case Name	Remove Deployment Service
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	User
Stakeholders and Interests	User: Remove the deployment service to stop site updates System Admin: Ensure removal does not cause system issues
Preconditions	User is logged in and has a connected deployment service
Success Guarantee (Postconditions)	Deployment service is disconnected from the project
Main Success Scenario	1. User navigates to Deployment Settings 2. System shows connected services 3. User selects service to disconnect 4. System confirms disconnection 5. System removes the service
Extensions	5a. User cancels disconnection: Service remains connected 6a. Service removal failure: System suggests retrying later
Special Requirements	Disconnection process <5 seconds, no active deployment during removal

Table 2.5: Remove Deployment Service

Use Case Name	Deploy Site
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	User
Stakeholders and Interests	User: Deploy the project to make it live Deployment Provider: Host the live website System Admin: Ensure deployment functions smoothly
Preconditions	User is logged in, project is ready for deployment, service is connected
Success Guarantee (Postconditions)	Project is successfully deployed and live
Main Success Scenario	1. User navigates to the Deploy option 2. System shows deployment settings 3. User initiates deployment 4. System processes and deploys the project 5. Site goes live and is accessible
Extensions	4a. Deployment fails: System shows error message and suggests retry 5a. Partial deployment: System highlights issues with certain components
Special Requirements	Deployment <10 seconds, rollback option available

Table 2.6: Deploy Site

Use Case Name	Export Code
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	User
Stakeholders and Interests	User: Export the project code for external use, backups, or further development System Admin: Ensure the code is production-ready
Preconditions	User is logged in with an active project
Success Guarantee (Postconditions)	Code is successfully exported and downloaded
Main Success Scenario	1. User selects Export Code option 2. System packages the project code 3. User downloads the export file
Extensions	3a. Code export fails: System shows error and suggests retry 4a. Partial export: System highlights issues with specific components
Special Requirements	Export process <5 seconds, code ready for use

Table 2.7: Export Code

Use Case Name	Track Analytics
Scope	Napalm Web Creation Kit
Level	System function
Primary Actor	System
Stakeholders and Interests	User: Track site usage and engagement metrics Admin: Monitor system performance and user engagement
Preconditions	The user has a deployed website
Success Guarantee (Postconditions)	Analytics data is tracked and stored successfully
Main Success Scenario	1. System tracks user interactions with the website 2. System stores analytics data 3. User or admin accesses the analytics reports
Extensions	2a. System fails to track data: System logs error and notifies admin
Special Requirements	Data collection should not impact site performance

Table 2.8: Track Analytics

Use Case Name	Run Tests
Scope	Napalm Web Creation Kit
Level	System function
Primary Actor	System
Stakeholders and Interests	User: Ensure project functionality through automated tests System Admin: Ensure the system meets performance and reliability standards
Preconditions	User has a project ready for testing
Success Guarantee (Postconditions)	All tests are run successfully and results are provided
Main Success Scenario	1. User triggers automated tests 2. System runs performance and functionality tests 3. System provides test results
Extensions	3a. Test failure: System displays error and suggests solution
Special Requirements	Test process <10 seconds for small projects

Table 2.9: Run Tests

Use Case Name	View Analytics
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	Admin
Stakeholders and Interests	Admin: Monitor advanced analytics for websites and projects System Admin: Ensure the accuracy and integrity of analytics data
Preconditions	Analytics data is collected for deployed websites
Success Guarantee (Postconditions)	Admin can view detailed analytics and make decisions
Main Success Scenario	1. Admin navigates to the Analytics dashboard 2. System displays traffic and usage data 3. Admin filters and reviews analytics data
Extensions	2a. System detects missing analytics data: System reports issue to admin
Special Requirements	Analytics reports should load within 3 seconds

Table 2.10: View Analytics

Use Case Name	Manage Users
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	Admin
Stakeholders and Interests	Admin: Manage user accounts, roles, and access permissions System Admin: Ensure secure handling of user data
Preconditions	Admin is logged in with appropriate permissions
Success Guarantee (Postconditions)	User accounts and roles are managed successfully
Main Success Scenario	1. Admin navigates to Manage Users 2. System displays user list 3. Admin adds, updates, or deletes users 4. System saves changes
Extensions	3a. Admin attempts to remove an active user: System prompts warning
Special Requirements	Secure handling of user data

Table 2.11: Manage Users

Use Case Name	Manage Templates
Scope	Napalm Web Creation Kit
Level	User goal
Primary Actor	Admin
Stakeholders and Interests	Admin: Manage templates used by users to create websites Users: Depend on templates for website creation
Preconditions	Admin is logged in with necessary permissions
Success Guarantee (Postconditions)	Templates are successfully created, updated, or removed
Main Success Scenario	1. Admin navigates to Manage Templates 2. System shows template list 3. Admin creates, updates, or removes templates 4. System saves changes
Extensions	3a. Invalid template structure: System warns admin 4a. Failure to save template: System suggests retry
Special Requirements	Templates should load quickly and be easy to edit

Table 2.12: Manage Templates

Chapter 3

System Overview

Napalm is a web-based platform designed to streamline website creation through a drag-and-drop interface, AI-powered content generation, and real-time collaboration. It enables users—both non-technical and advanced—to create professional websites with minimal effort, while still allowing for code customization. The system supports various functionalities such as user authentication, deployment to third-party services, and secure data management.

Napalm utilizes modern web development technologies such as Next.js, MongoDB, and WebSockets to offer both a seamless user experience and backend scalability. This section provides a detailed breakdown of the architecture, design models, and data structures used in the project.

3.1 Architectural Design

Napalm employs a hybrid architectural pattern combining Client-Server and Model-View-Controller (MVC) approaches. This dual architecture allows for optimal separation of concerns while leveraging the benefits of modern web technologies. The Client-Server pattern facilitates network communication between users' browsers and our Next.js server, while the MVC pattern organizes the codebase for maintainability and scalability.

The system is divided into six primary modules, with Modules 1 (UI Design) and 2 (AI Content Generation) running on the client-side, Module 3 (User Authentication and Database) operating server-side, Module 4 (WebSocket) bridging client-server communication, Module 5 handling external deployments, and Module 6 supporting development optimization.

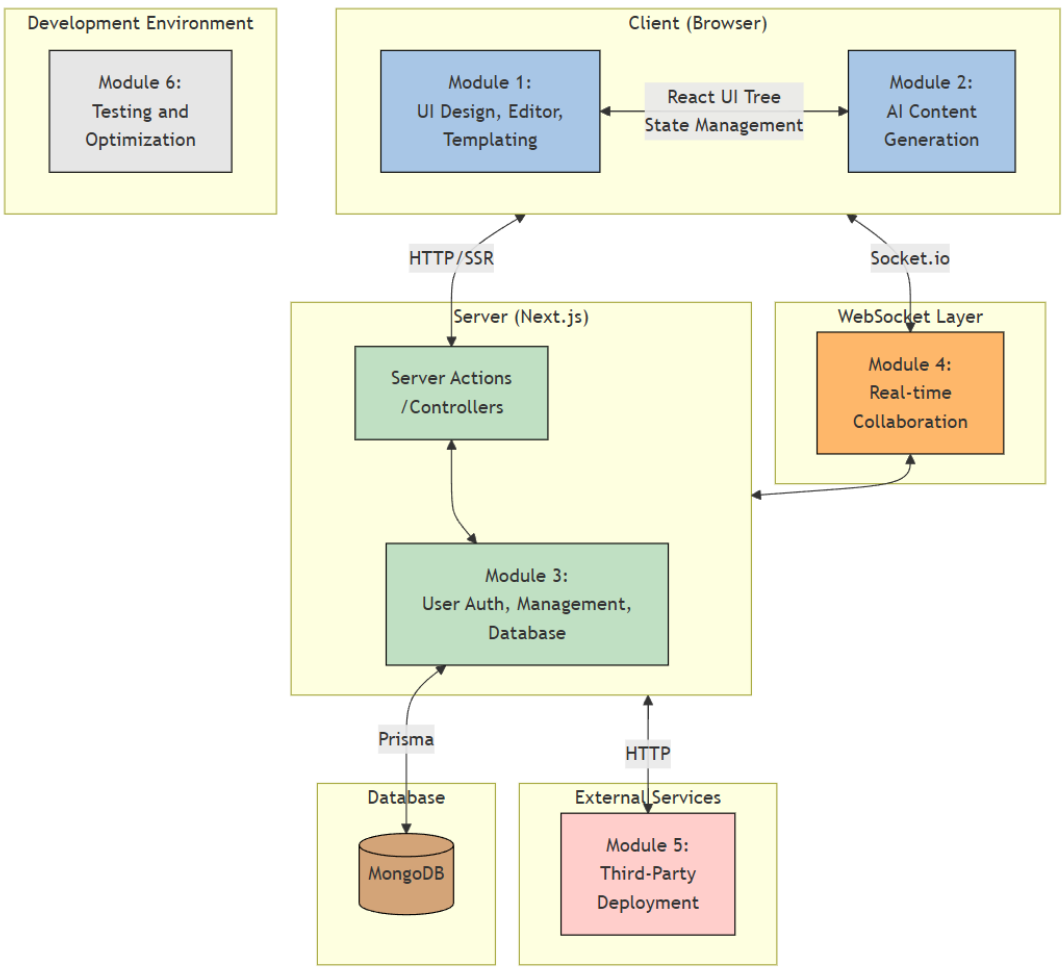


Figure 3.1: Box and Line Diagram

The box and line diagram illustrates the high-level system architecture. On the client side, the UI Design and AI Content Generation modules interact through React's component tree and state management. The WebSocket layer enables real-time collaboration between clients and the server. Server-side components handle data persistence and authentication through MongoDB and Prisma, while external services are integrated for deployment capabilities.

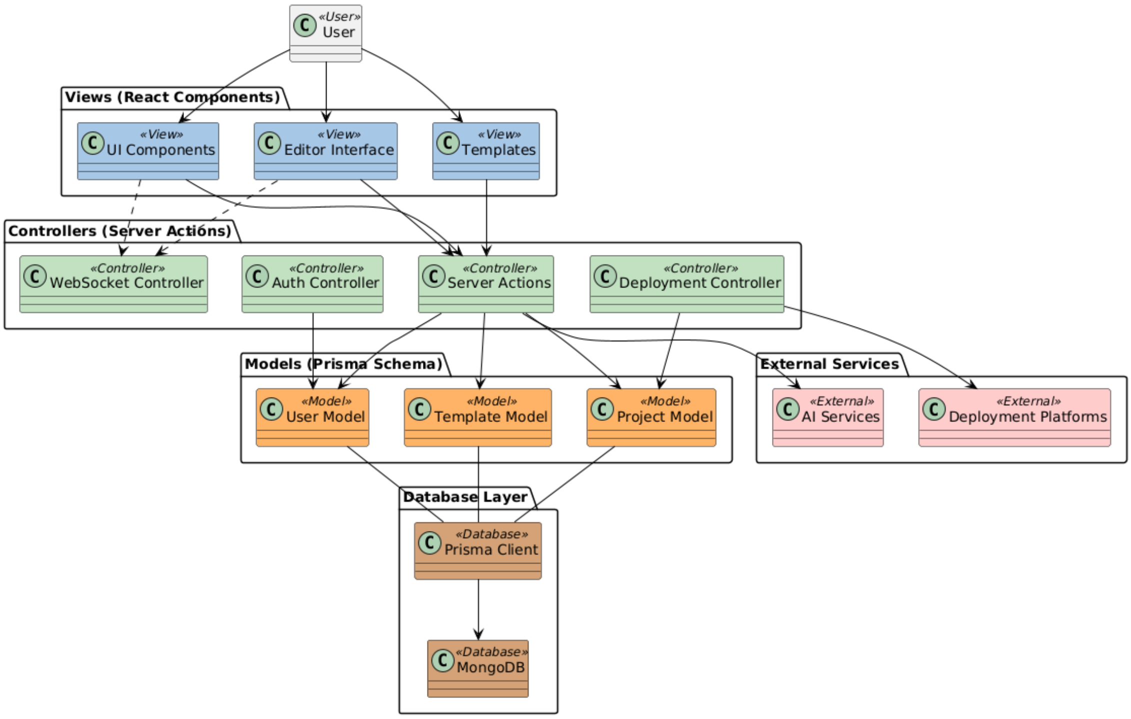


Figure 3.2: MVC Pattern Diagram

The MVC pattern diagram details how Napalm implements separation of concerns: - Models are defined using Prisma schema, representing data structures for users, projects, and templates - Views are React components, creating the user interface for the editor and templates - Controllers are implemented as Next.js server actions, handling business logic and data flow

This architecture ensures scalability, maintainability, and a clear separation of concerns while enabling the real-time collaborative features essential to Napalm's functionality.

3.2 Design Models

Napalm follows an object-oriented development approach, where design models help describe the flow and structure of the system. The following models are crucial for understanding the behavior of various system components:

3.2.1 Activity Diagrams

3.2.1.1 Login

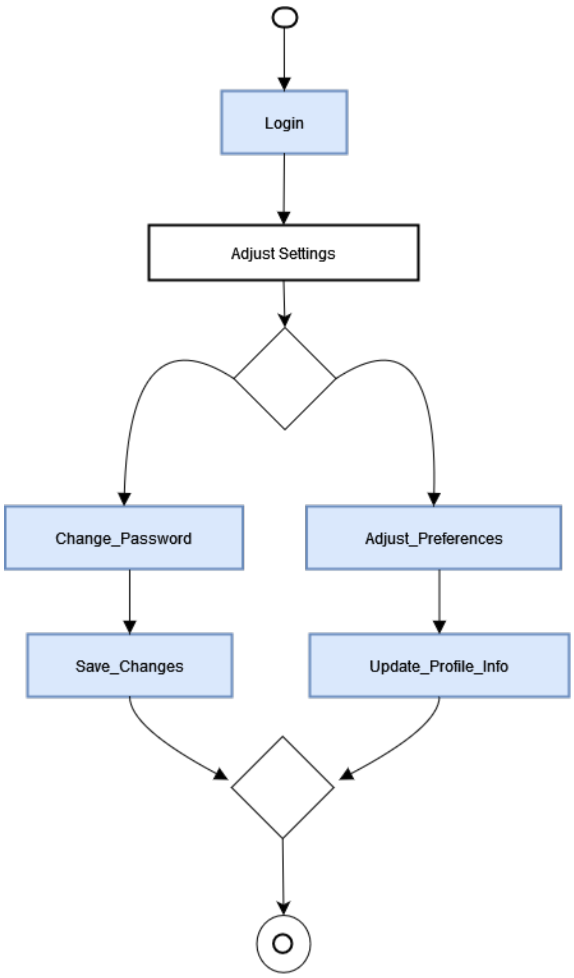


Figure 3.3: Login Activity

3.2.1.2 Manage Projects

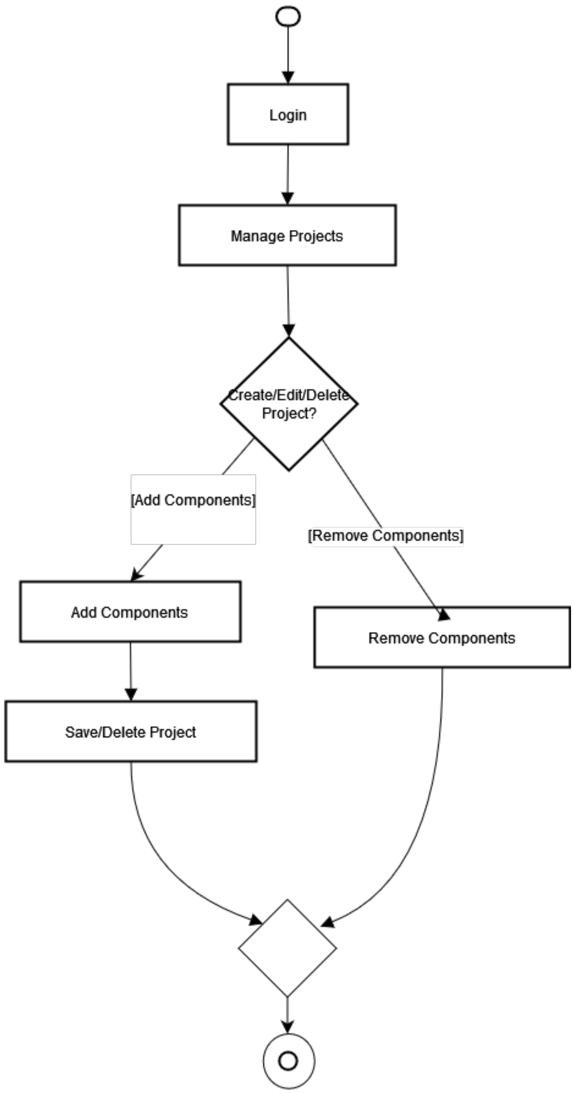


Figure 3.4: Manage Projects Activity

3.2.1.3 Prompt AI

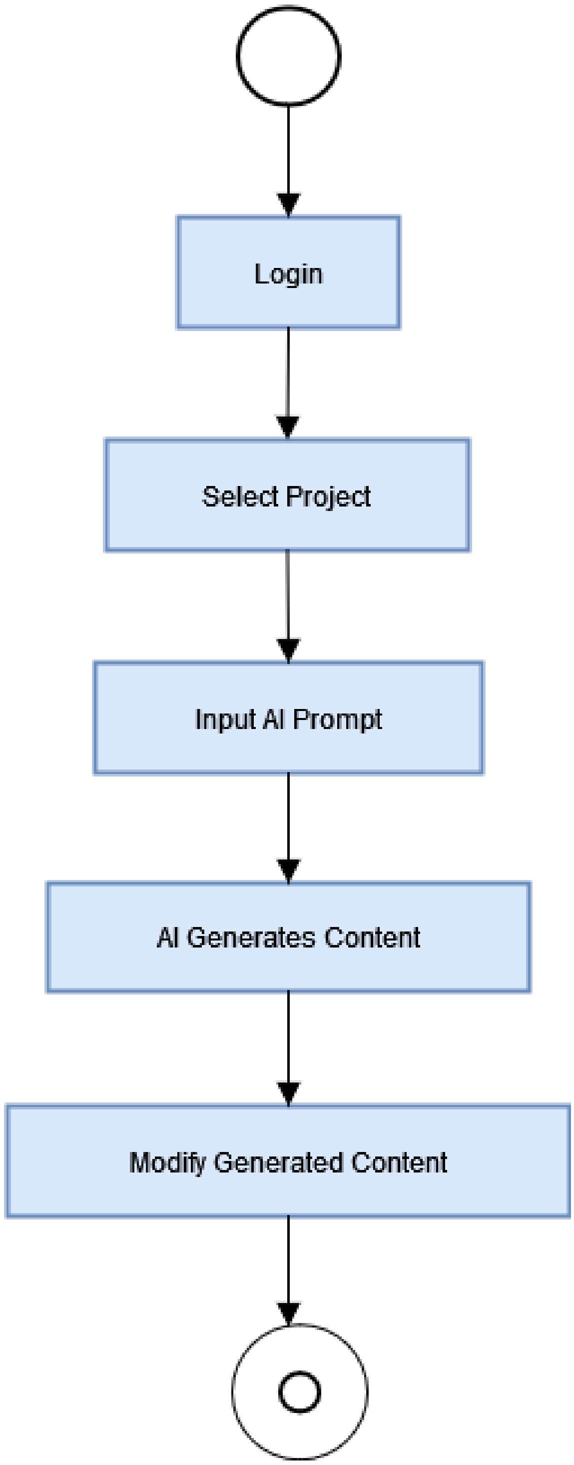


Figure 3.5: Prompt AI Activity

3.2.1.4 Connect Deployment

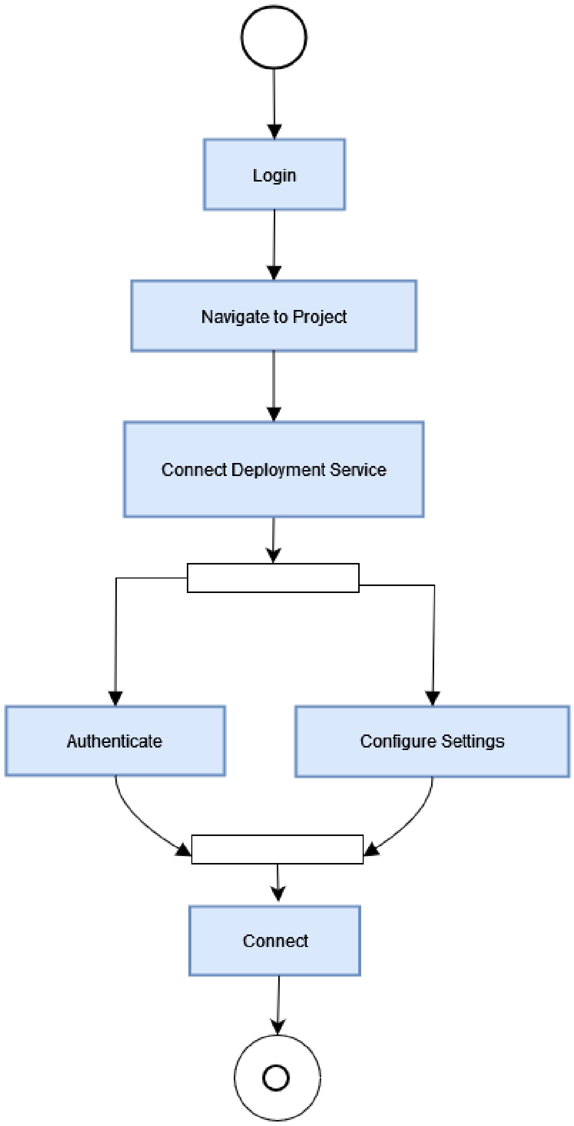


Figure 3.6: Connect Deployment Activity

3.2.1.5 Remove Deployment

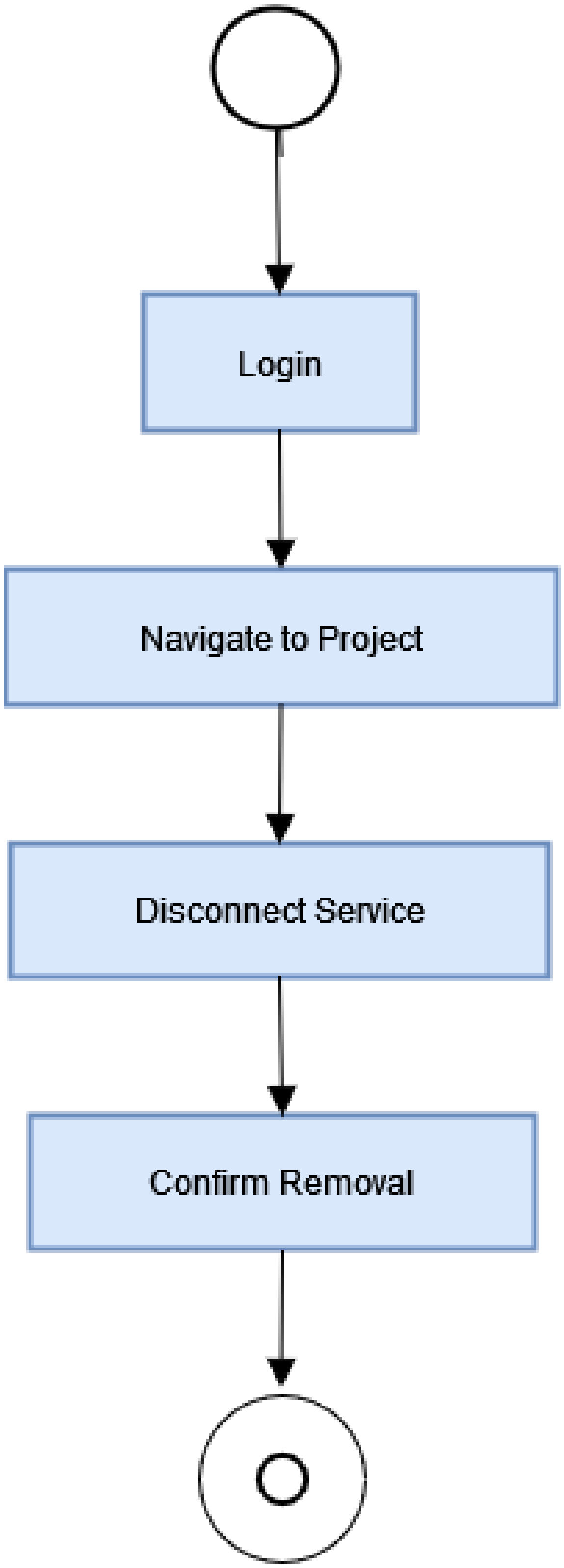


Figure 3.7: Remove Deployment Activity

3.2.1.6 Deploy Site

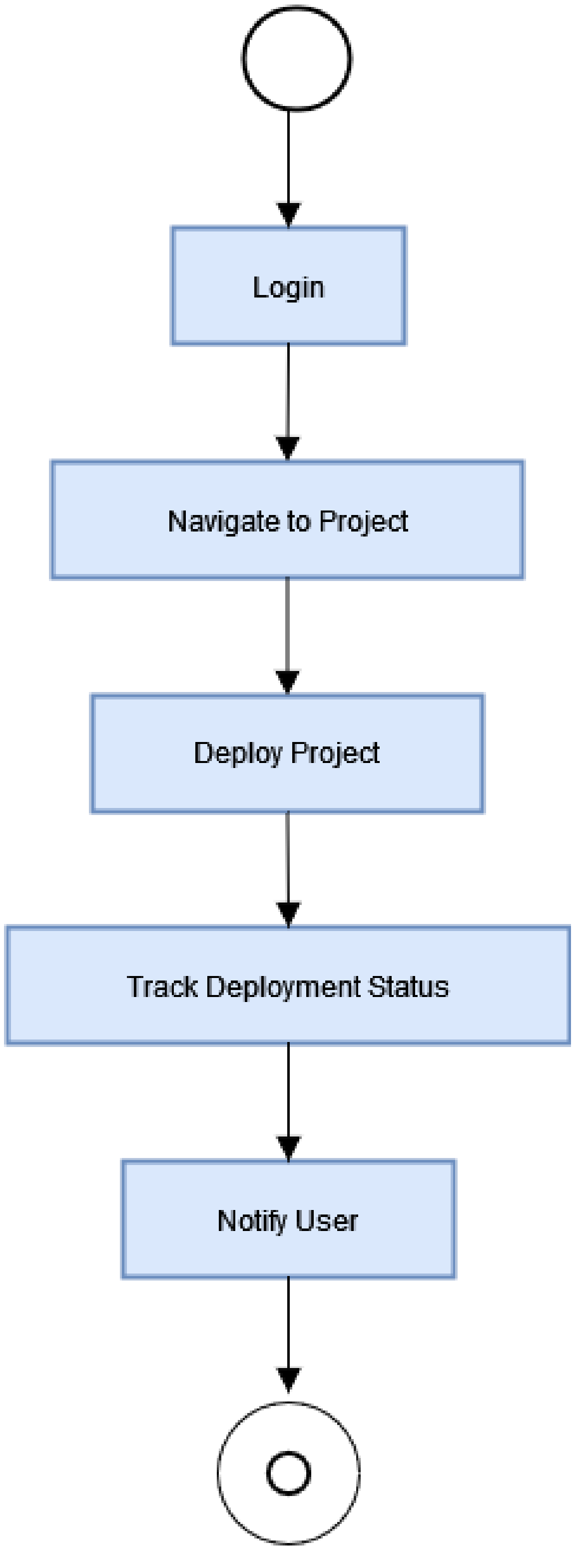
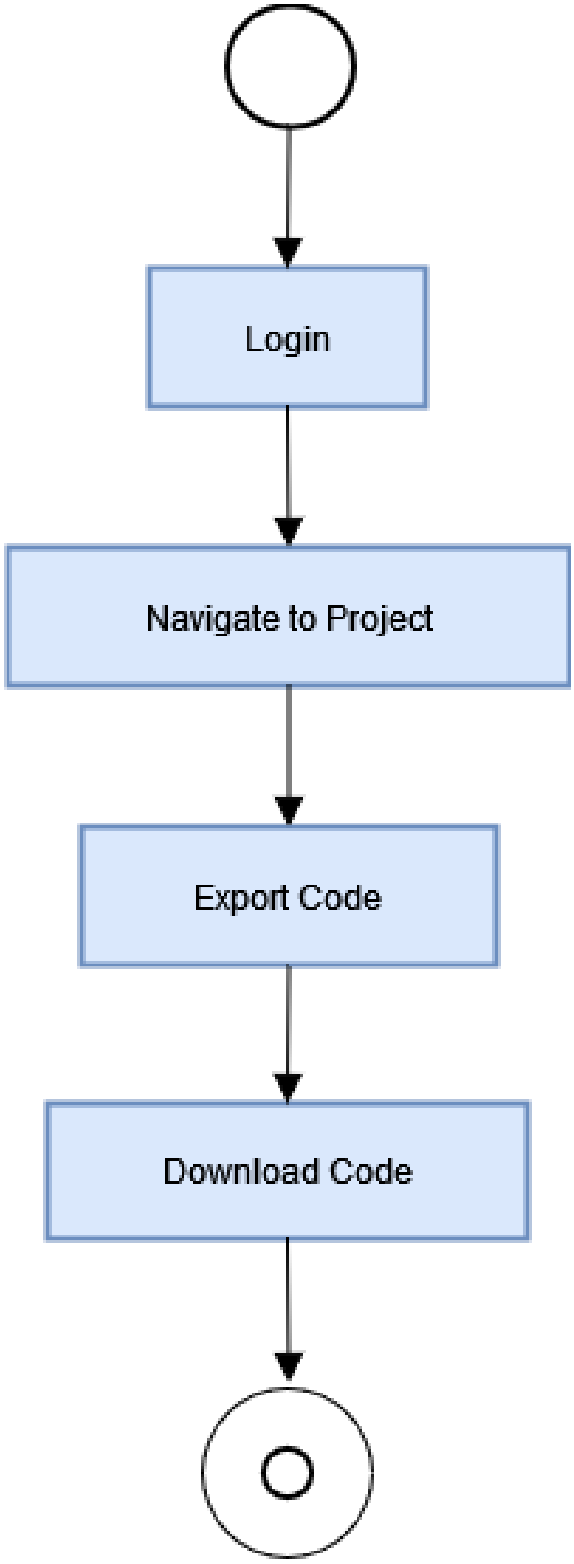


Figure 3.8: Deploy Site Activity

3.2.1.7 Export Code



3.2.1.8 Track Analytics

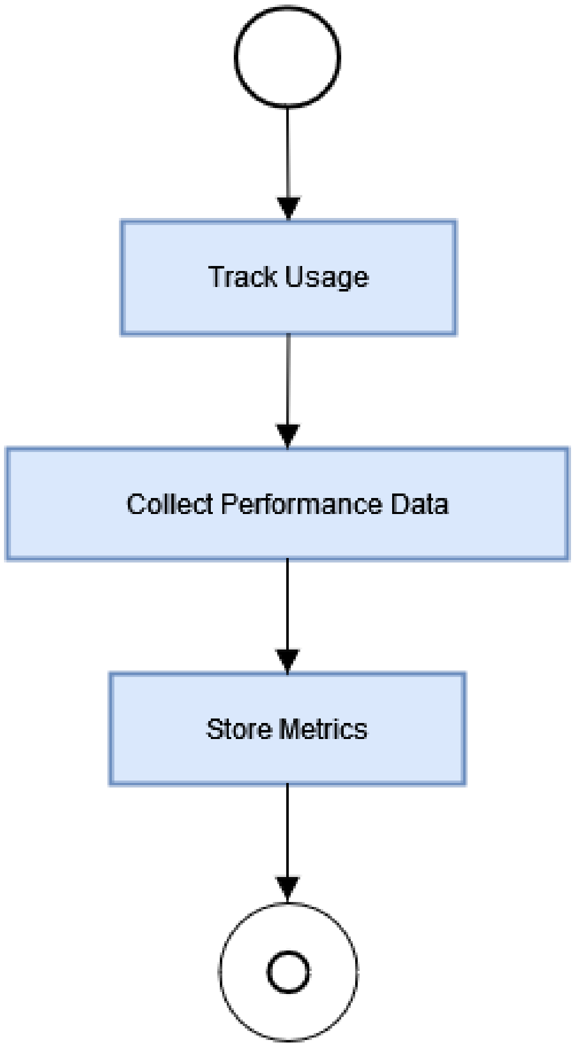


Figure 3.10: Track Analytics Activity

3.2.1.9 Run Tests

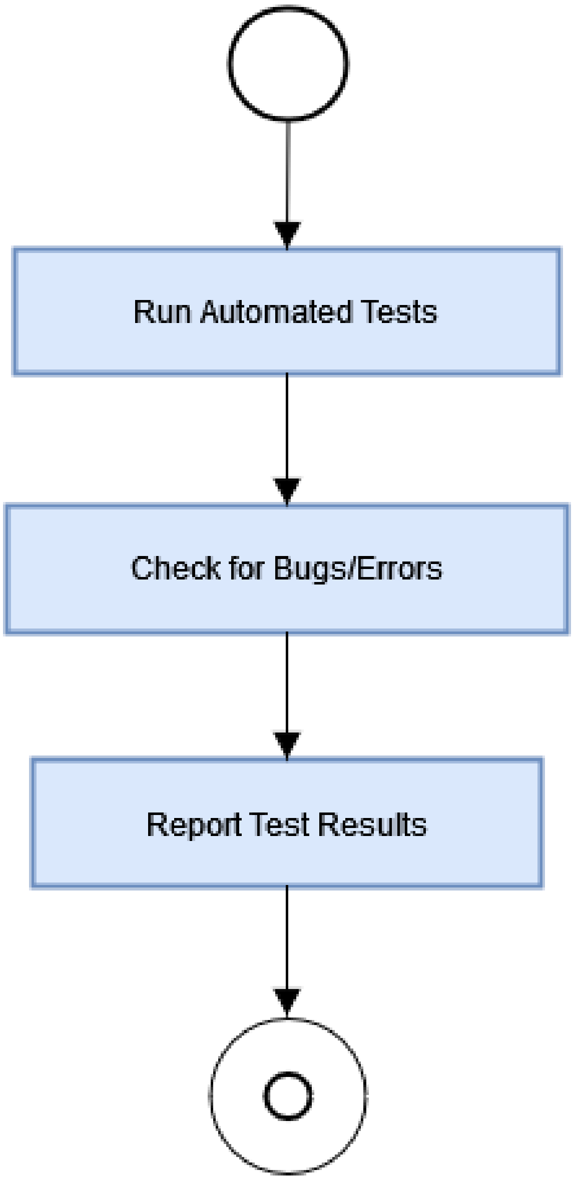


Figure 3.11: Run Tests Activity

3.2.1.10 View Analytic

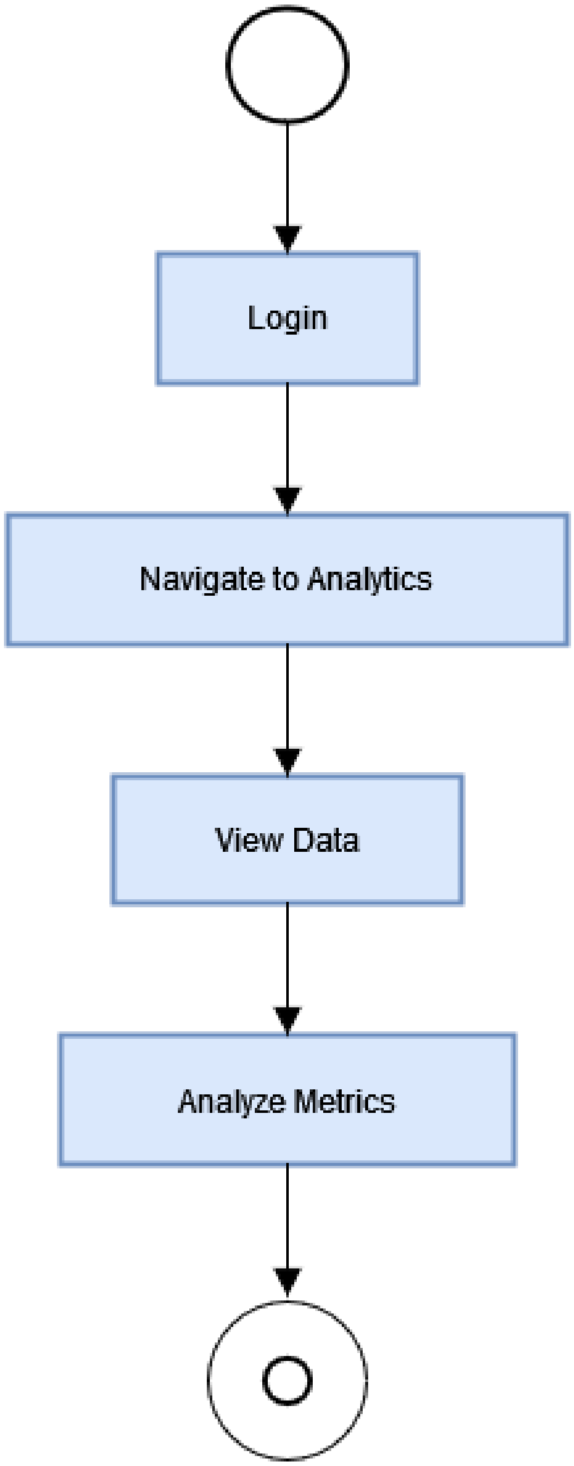


Figure 3.12: View Analytic Activity

3.2.1.11 Manage Users

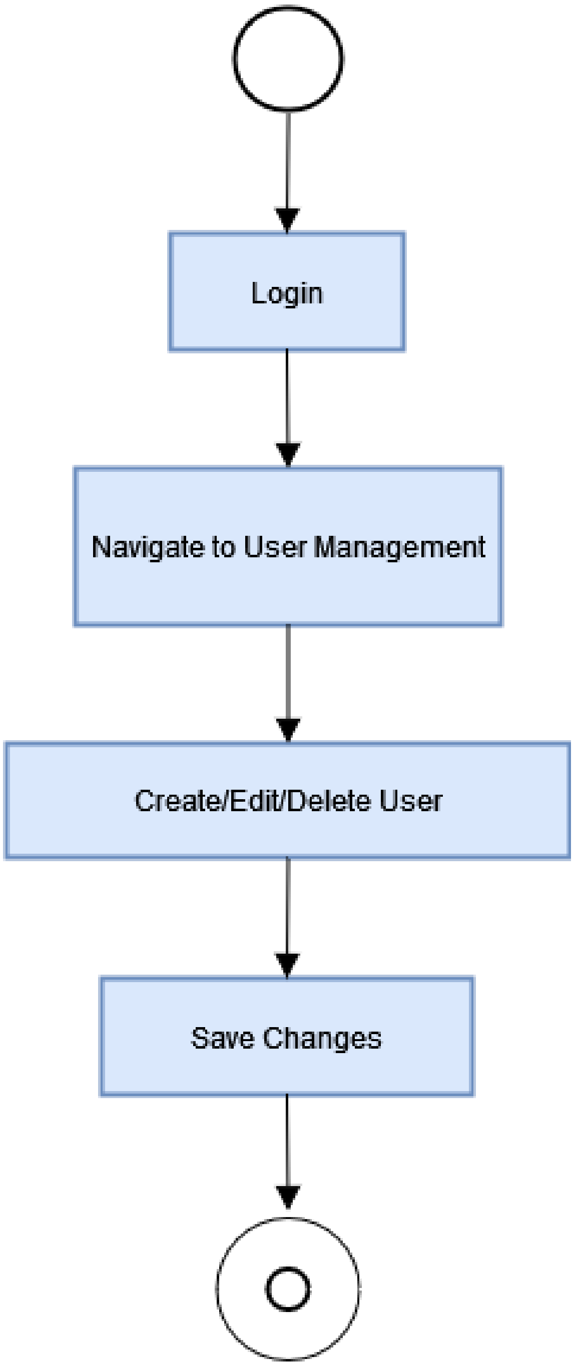
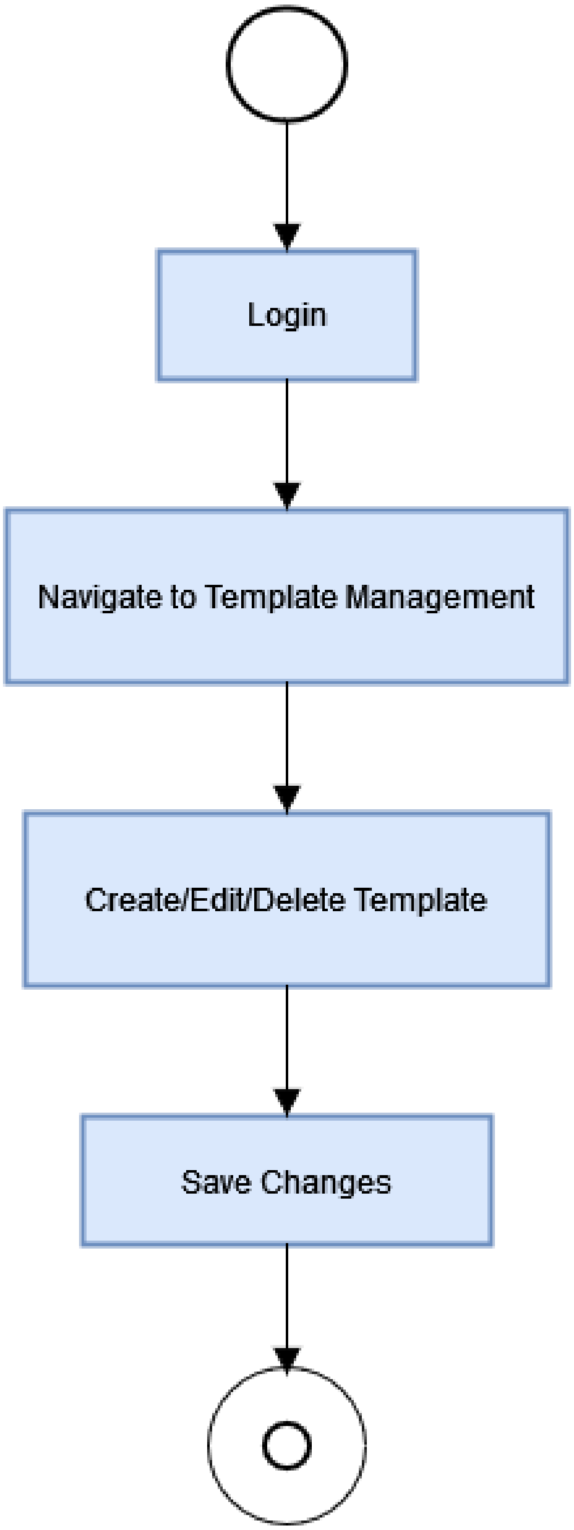


Figure 3.13: Manage Users Activity

3.2.1.12 Manage Templates



3.2.2 Sequence Diagrams

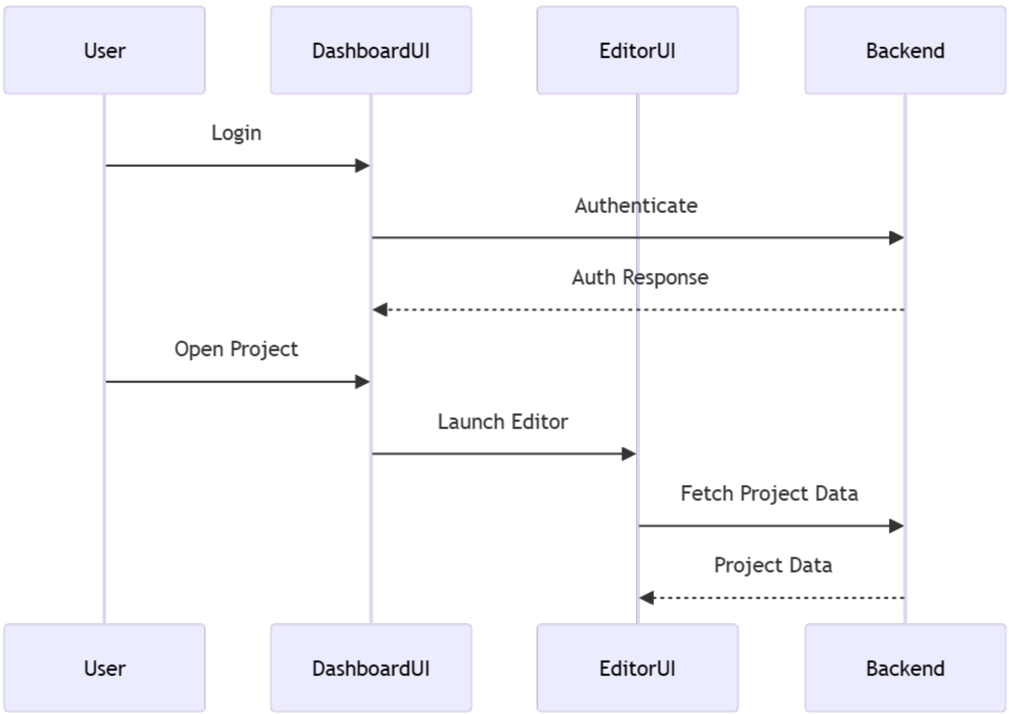


Figure 3.15: Authentication and Project Opening

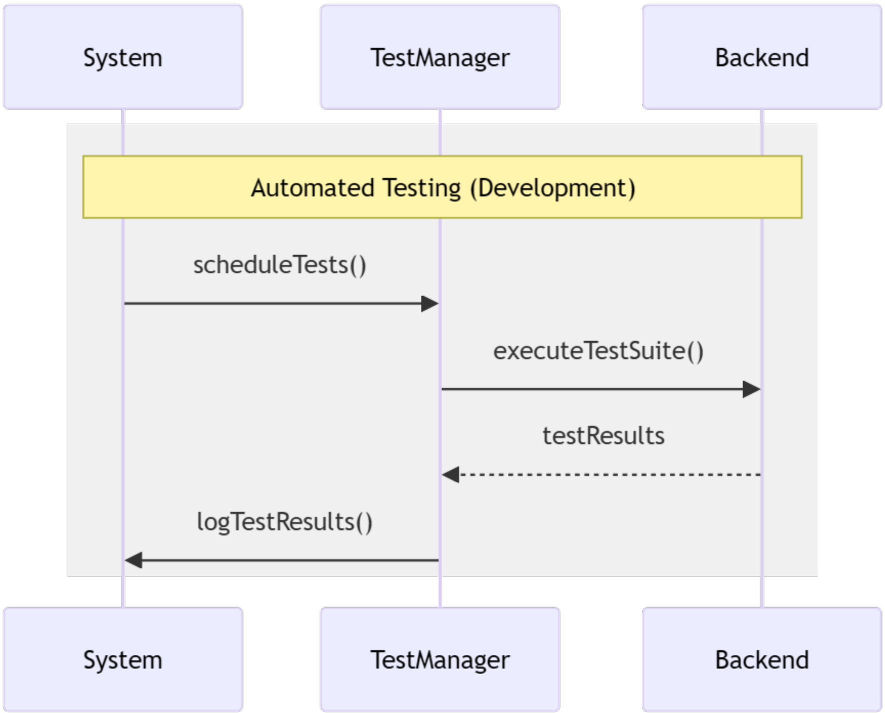


Figure 3.16: Automated Testing

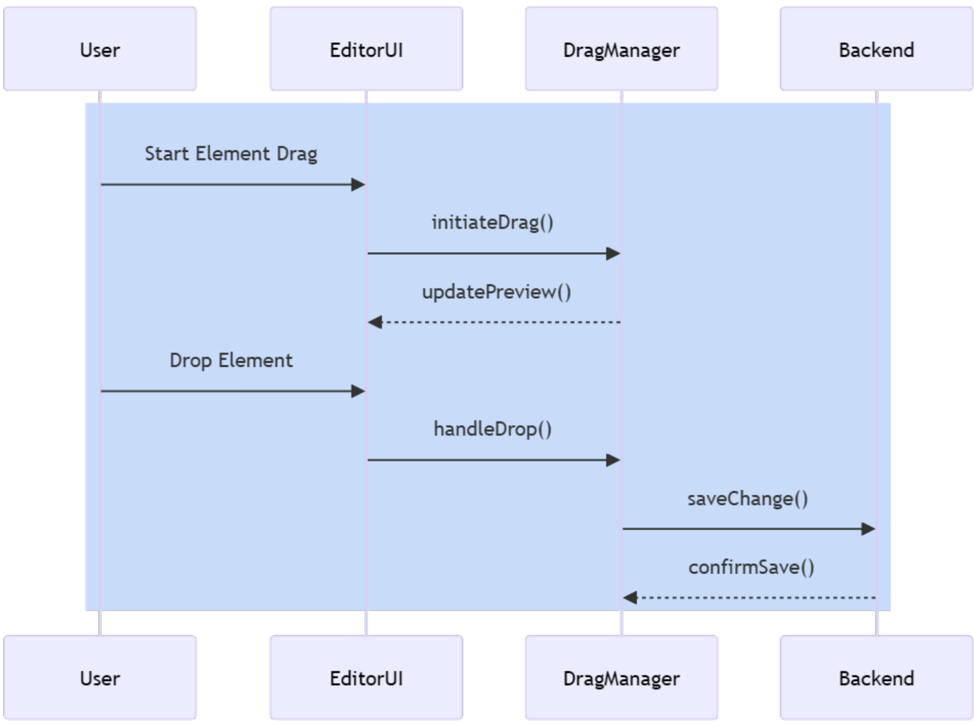


Figure 3.17: Editor Interactions

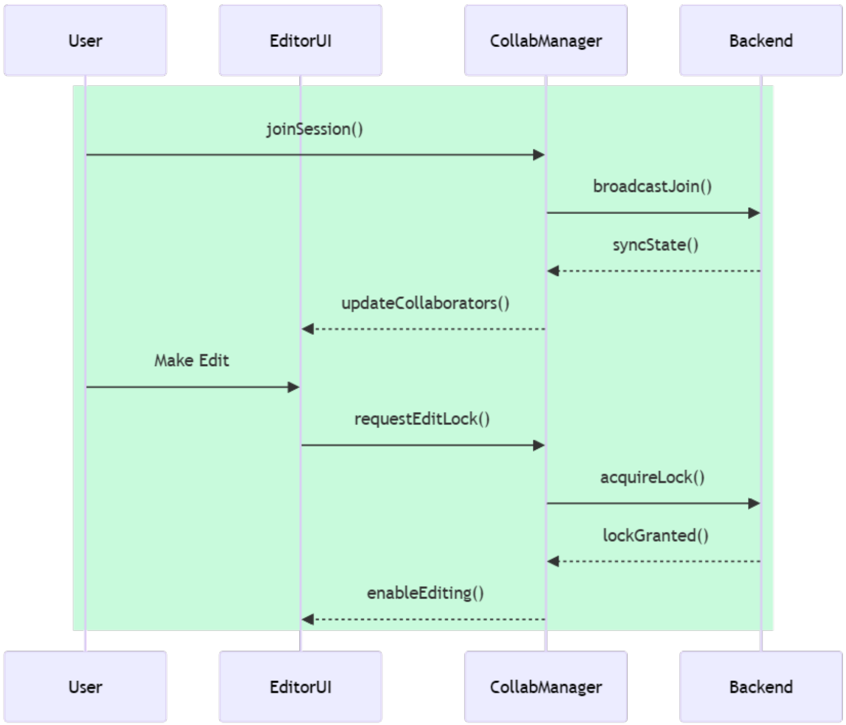


Figure 3.19: Real-time Collaboration

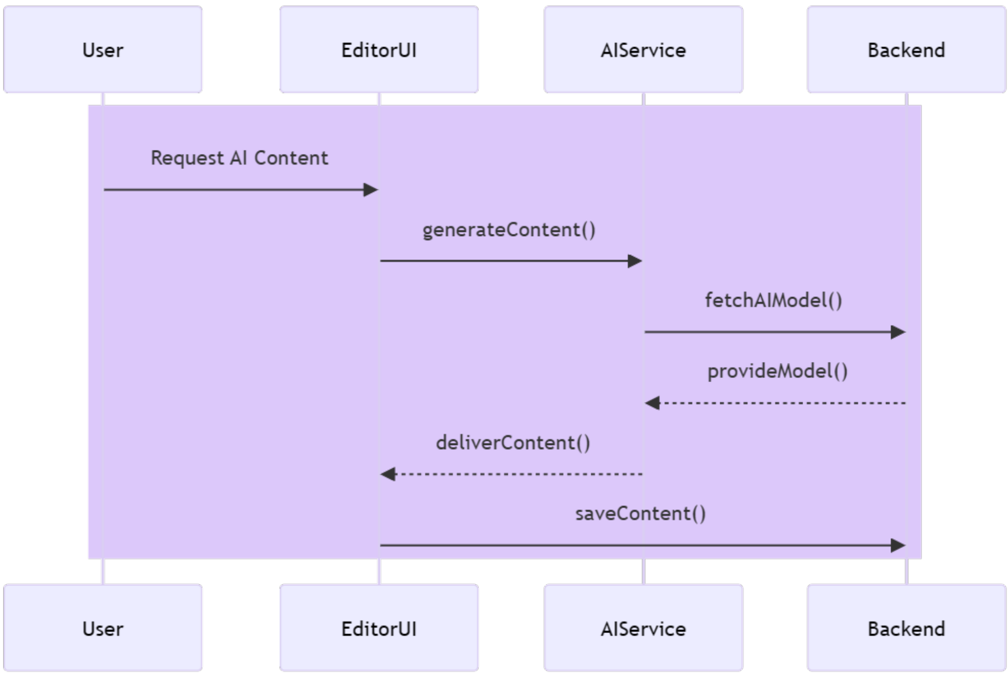


Figure 3.18: AI Content Generation

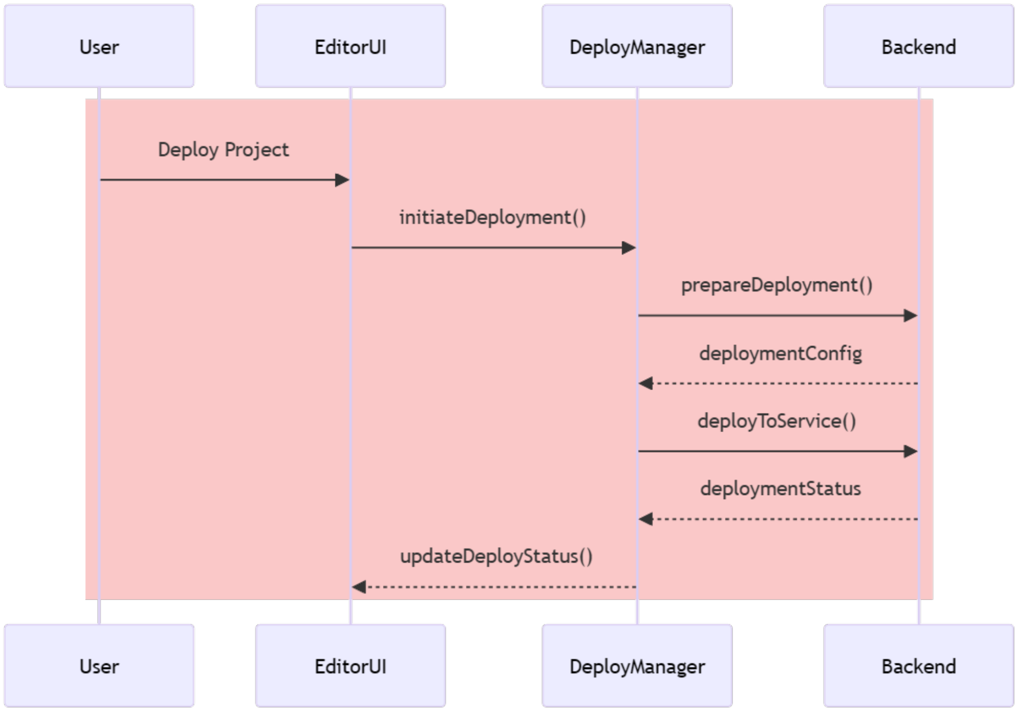


Figure 3.20: Deployment Process

3.2.3 Class Diagram

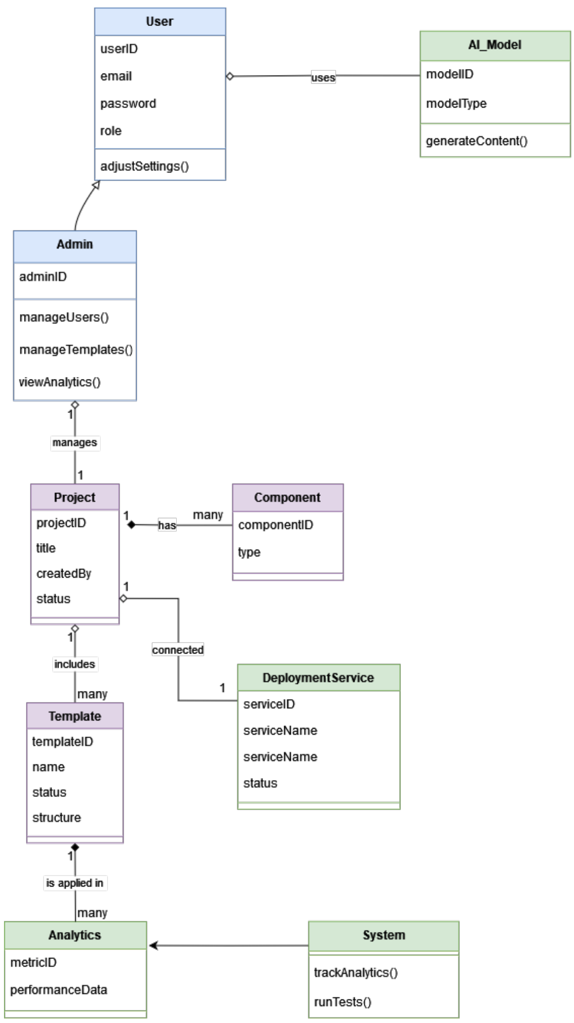


Figure 3.21: Class Diagram

3.2.4 State Transition Diagram

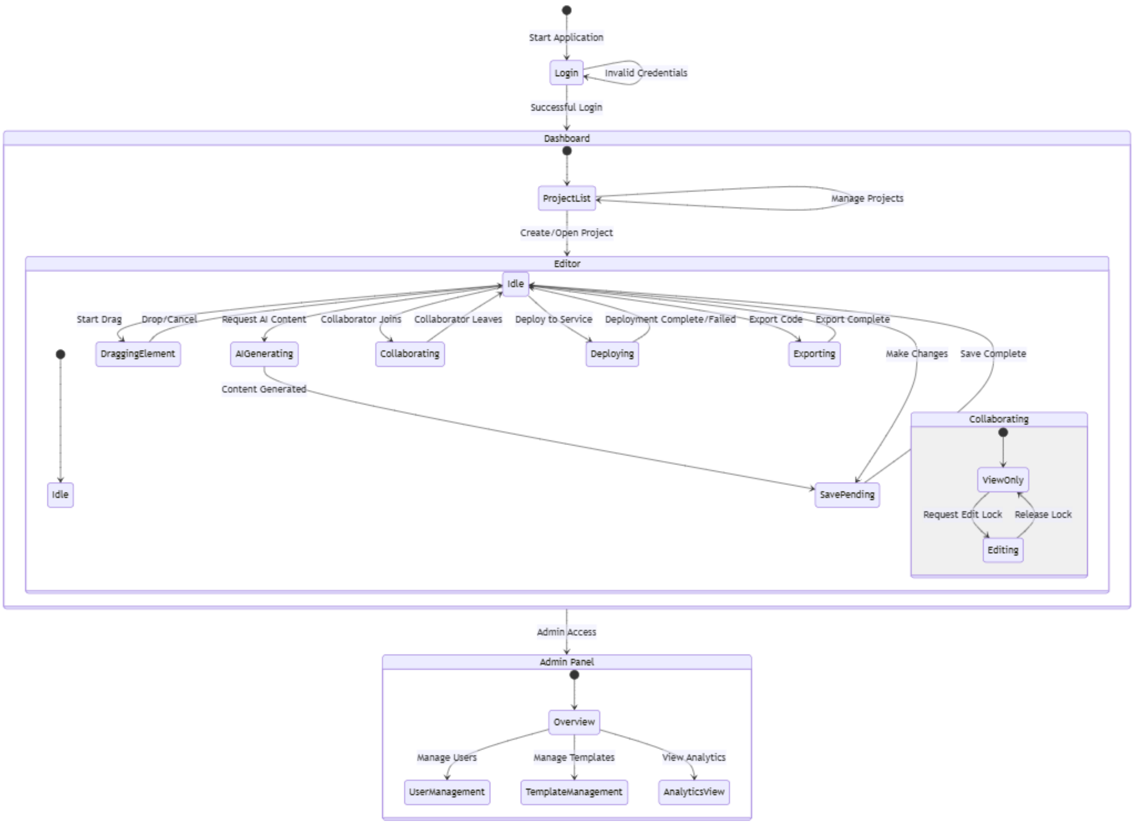


Figure 3.22: State Transition Diagram

3.3 Data Design

Data design focuses on how information within Napalm is structured, stored, and retrieved. The system utilizes MongoDB as the primary database, with Prisma as the ORM (Object-Relational Mapping) tool to manage data interaction efficiently.

3.3.1 Data Entities

• User Data: Stored in the database with fields such as userID, email, hashed password, role (admin or normal user), and profile information.

• Project Data: Each project created by a user has an associated projectID, title, description, and links to templates or customized content.

• Template Data: Templates are stored with identifiers like templateID, template name, and layout structure (JSON format).

• Component Data: Components in the drag-and-drop editor are saved in a format that preserves their properties, layout, and hierarchy in the user's project.

3.3.2 Data Flow

When a user creates or edits a project:

• The system captures the project state and stores it in MongoDB.

• The user can interact with templates and AI models to generate content. All interactions are saved and synchronized in real time.

• When deploying, the system exports the project as production-ready code, which can then be deployed to Vercel or Netlify.

3.3.3 Data Integrity and Security

Napalm implements stringent data security measures, including:

• Encryption: Sensitive data such as passwords are encrypted using secure hashing algorithms.

• Backup: Data is backed up regularly to prevent data loss.

• Access Control: Different levels of access are provided for normal users and admins, ensuring data privacy.

3.4 References

1. Wix Support Article [3]
2. Forbes Advisor Article [1]
3. Seattle New Media Article [2]

Bibliography

[1] Forbes Advisor. Website statistics, 2023. Accessed: 2024-10-05.

[2] Seattle New Media. Why no-code/low-code is the future of development, 2023. Accessed: 2024-10-05.

[3] Wix Support. Top reasons for choosing wix, n.d. Accessed: 2024-10-05.